electricity

All 370 clients. Portuguese civil labels remain unresolved local times: DST spring zeros, autumn aggregated hours and pre-creation zero padding are retained, with no pristine-instant or outage assumption.

Completed evidence · 1 sources · 41 retained records

9 evidence-qualified finding(s) across 1 supplied source(s). Required stages completed over eligible records; behavioral search scope is disclosed separately.

Actions to investigate

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t8 · MT_001

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t12 · MT_002

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t16 · MT_003

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t20 · MT_004

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t24 · MT_005

Interpretation boundary

- No composite score or universal readiness verdict.
- Investigation defaults are not validated defect criteria.
- Supplied evidence does not prove what a production pipeline consumed.

Generated 2026-10-10T21:27:47.495661+00:00 · Schema 1.4 · temporal_reliability_v1

Declarations

Evidence / source	Measurement or rule	Retained result
electricity	Source and base task	{"format":"parquet","role":"dataset","task":"dataset_profile","targets":null}
electricity	observation	measurement
electricity	layout	wide
electricity	entity columns	[]
electricity	observation keys	[]
electricity	clock	{"column":"time","kind":"local_datetime","encoding":"native","format":null,"epoch_unit":null,"timezone":null,"timezone_column":null,"ambiguity":"unresolved","fold_column":null,"date_convention":null}
electricity	schedule	{"kind":"none","cadence_ns":null,"anchor":null,"civil_unit":null,"timezone":null,"month_end":"skip","source_id":null,"time_column":null,"entity_columns":[],"measurement_column":null,"expectation":"required_record","collection_status_column":null,"collection_active_values":["active"]}
electricity	active intervals	[]
electricity	MT_001	{"name":"MT_001","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_002	{"name":"MT_002","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_003	{"name":"MT_003","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_004	{"name":"MT_004","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_005	{"name":"MT_005","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_006	{"name":"MT_006","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_007	{"name":"MT_007","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_008	{"name":"MT_008","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_009	{"name":"MT_009","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_010	{"name":"MT_010","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_011	{"name":"MT_011","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}
electricity	MT_012	{"name":"MT_012","column":null,"representation":"quantitative","unit":"kW","meaning":null,"past_observation":false}

20 of 378 metric rows shown. 358 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

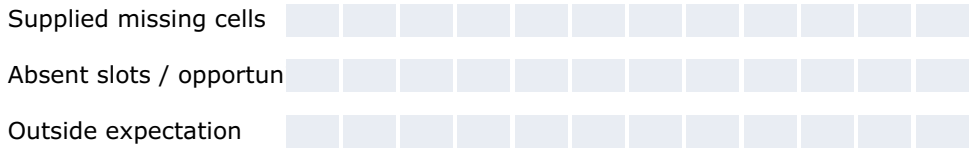
Chronology

Evidence / source	Measurement or rule	Retained result
t1 electricity	rows	140256
t1 electricity	cells	51894720
t1 electricity	declared dense missing cells	0
t1 electricity	parse states / valid	51894720
t1 electricity	precision counts / ms	51894720
t1 electricity	value states [MT_001] / measurement	MT_001
t1 electricity	value states [MT_001] / state	valid
t1 electricity	value states [MT_001] / cells	140256
t1 electricity	value states [MT_002] / measurement	MT_002
t1 electricity	value states [MT_002] / state	valid
t1 electricity	value states [MT_002] / cells	140256
t1 electricity	value states [MT_003] / measurement	MT_003
t1 electricity	value states [MT_003] / state	valid
t1 electricity	value states [MT_003] / cells	140256
t1 electricity	value states [MT_004] / measurement	MT_004
t1 electricity	value states [MT_004] / state	valid
t1 electricity	value states [MT_004] / cells	140256
t1 electricity	value states [MT_005] / measurement	MT_005
t1 electricity	value states [MT_005] / state	valid
t1 electricity	value states [MT_005] / cells	140256

20 of 1131 metric rows shown. 1111 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Continuity and history

Absent observations, supplied missing values and outside-expectation periods have different meanings. Event absence alone does not prove an outage.



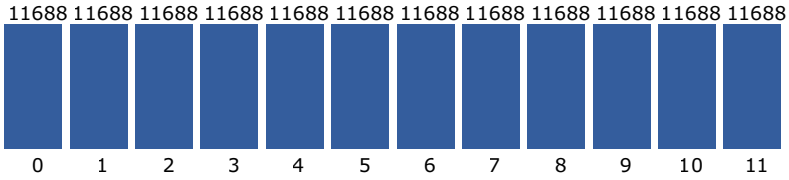
electricity · missing values and absent slots remain separate

Evidence / source	Measurement or rule	Retained result
t5 electricity	distinct entities	1
t5 electricity	distinct series	370
t5 electricity	history counts / min	140256
t5 electricity	history counts / max	140256
t5 electricity	history counts / entity weighted mean	140256
t5 electricity	history length bins [0] / observations	4
t5 electricity	history length bins [0] / series	370
t5 electricity	observed first tick	1293840900000000000
t5 electricity	observed last tick	1420070400000000000
t5 electricity	observed entry exit is lifecycle	No
t5 electricity	retained histories [MT_001] / entity	[]
t5 electricity	retained histories [MT_001] / measurement	MT_001
t5 electricity	retained histories [MT_001] / observations	140256
t5 electricity	retained histories [MT_001] / cells	140256
t5 electricity	retained histories [MT_001] / first tick	1293840900000000000
t5 electricity	retained histories [MT_001] / last tick	1420070400000000000
t5 electricity	retained histories [MT_001] / span ticks	126229500000000000
t5 electricity	retained histories [MT_002] / entity	[]
t5 electricity	retained histories [MT_002] / measurement	MT_002
t5 electricity	retained histories [MT_002] / observations	140256

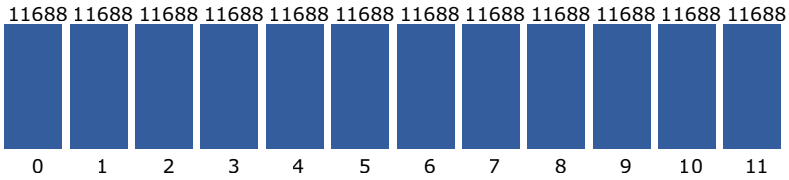
20 of 836 metric rows shown. 816 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Behavioral investigation

Thresholds and support settings are configurable investigation defaults. Reference support, sparse windows, zero dispersion, phase exclusions and population mix qualify interpretation.



electricity / MT_001 · supplied cells per window



electricity / MT_002 · supplied cells per window

Evidence / source	Measurement or rule	Retained result
t8 electricity / MT_001	Reference and interpretation	{"reference_window":0,"unit":"kW","meaning":"Observed effects are descriptive investigation evidence, not validated universal defects, changepoints, concept drift, or model performance."}
t8 electricity / MT_001	Window 0	{"window_id":0,"start_tick":"1293840900000000000","end_tick":"1304360025000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0],"iqr":0.0}
t8 electricity / MT_001	Window 1	{"window_id":1,"start_tick":"1304360025000000000","end_tick":"1314879150000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0],"iqr":0.0}
t8 electricity / MT_001	Window 2	{"window_id":2,"start_tick":"1314879150000000000","end_tick":"1325398275000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0],"iqr":0.0}
t8 electricity / MT_001	Window 3	{"window_id":3,"start_tick":"1325398275000000000","end_tick":"1335917400000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[1.26903553299492,1.26903553299492,3.80710659898477,7.61421319796954,17.7664974619289,20.3045685279188,34.2639593908629],"iqr":13.95939086294413}
t8 electricity / MT_001	Window 4	{"window_id":4,"start_tick":"1335917400000000000","end_tick":"1346436525000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[1.26903553299492,1.26903553299492,1.26903553299492,2.53807106598985,3.80710659898477,16.497461928934,19.035532994923905],"iqr":2.53807106598985}
t8 electricity / MT_001	Window 5	{"window_id":5,"start_tick":"1346436525000000000","end_tick":"1356955650000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[1.26903553299492,1.26903553299492,1.26903553299492,2.53807106598985,2.53807106598985,3.80710659898477,5.076142131979 7],"iqr":1.26903553299493}
t8 electricity / MT_001	Window 6	{"window_id":6,"start_tick":"1356955650000000000","end_tick":"1367474775000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[1.26903553299492,1.26903553299492,1.26903553299492,2.53807106598985,2.53807106598985,5.0761421319797,16.497461928934],"iqr":1.26903553299493}
t8 electricity / MT_001	Window 7	{"window_id":7,"start_tick":"1367474775000000000","end_tick":"1377993900000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,1.26903553299492,2.53807106598985,10.1522842639594,16.497461928934,24.1116751269036],"iqr":8.88324873096448}

Evidence / source	Measurement or rule	Retained result
t8 electricity / MT_001	Window 8	{"window_id":8,"start_tick":"1377993900000000000","end_tick":"1388513025000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,1.26903553299492,2.53807106598985,13.9593908629442,16.497461928934,17.7664974619289,32.994923857868],"iqr":13.95939086294415}
t8 electricity / MT_001	Window 9	{"window_id":9,"start_tick":"1388513025000000000","end_tick":"1399032150000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,1.26903553299492,2.53807106598985,2.53807106598985,5.0761421319797,16.497461928934],"iqr":1.26903553299493}
t8 electricity / MT_001	Window 10	{"window_id":10,"start_tick":"1399032150000000000","end_tick":"1409551275000000000","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,1.26903553299492,2.53807106598985,15.2284263959391,16.497461928934,27.9187817258883],"iqr":13.959390862944181}
t8 electricity / MT_001	Window 11	{"window_id":11,"start_tick":"1409551275000000000","end_tick":"1420070400000000001","cells":11688,"missing":0,"invalid":0,"nonfinite":0,"finite":11688,"entities":1,"missing_rate":0.0,"quantiles":[0.0,0.0,1.26903553299492,2.53807106598985,2.53807106598985,16.497461928934,17.7664974619289],"iqr":1.26903553299493}
t8 electricity / MT_001	Comparison 0 → 1	{"reference_window":0,"comparison_window":1,"reference_support":11688,"comparison_support":11688,"reference_iqr":0.0,"reference_span":["1293840900000000000","1304360025000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_distan… [retained JSON contains full detail]
t8 electricity / MT_001	Comparison 1 → 2	{"reference_window":1,"comparison_window":2,"reference_support":11688,"comparison_support":11688,"reference_iqr":0.0,"reference_span":["1304360025000000000","1314879150000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_distan… [retained JSON contains full detail]
t8 electricity / MT_001	Comparison 2 → 3	{"reference_window":2,"comparison_window":3,"reference_support":11688,"comparison_support":11688,"reference_iqr":0.0,"reference_span":["1314879150000000000","1325398275000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_distan… [retained JSON contains full detail]
t8 electricity / MT_001	Comparison 3 → 4	{"reference_window":3,"comparison_window":4,"reference_support":11688,"comparison_support":11688,"reference_iqr":13.95939086294413,"reference_span":["1325398275000000000","1335917400000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":fal… [retained JSON contains full detail]
t8 electricity / MT_001	Comparison 4 → 5	{"reference_window":4,"comparison_window":5,"reference_support":11688,"comparison_support":11688,"reference_iqr":2.538071065989855,"reference_span":["1335917400000000000","1346436525000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":fals… [retained JSON contains full detail]
t8 electricity / MT_001	Comparison 5 → 6	{"reference_window":5,"comparison_window":6,"reference_support":11688,"comparison_support":11688,"reference_iqr":1.269035532994943,"reference_span":["1346436525000000000","1356955650000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":fals… [retained JSON contains full detail]
t8 electricity / MT_001	Comparison 6 → 7	{"reference_window":6,"comparison_window":7,"reference_support":11688,"comparison_support":11688,"reference_iqr":1.269035532994943,"reference_span":["1356955650000000000","1367474775000000000"],"reference_entities":1,"comparison_entities":1,"unit":"kW","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":fals… [retained JSON contains full detail]

20 of 5126 metric rows shown. 5106 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 6 / 8

Availability and outcome maturity

Undeclared availability remains unknown. Historical agreement concerns supplied coverage; it does not establish what a pipeline consumed.

Evidence / source	Measurement or rule	Retained result
t7 electricity	features	["MT_001","MT_002","MT_003","MT_004","MT_005","MT_006","MT_007","MT_008","MT_009","MT_010","MT_011","MT_012","MT_013","MT_014","MT_015","MT_016","MT_017","MT_018","MT_019","MT_020","MT_021","MT_022","MT_023","MT_024","MT_025","MT_026","MT_027","MT_028","MT_029","MT_030","MT_031","MT_032","MT_033","MT_034","MT_035","MT_036","MT_037","MT_038","MT_… [retained JSON contains full detail]
t7 electricity	meaning	Undeclared availability remains unknown; no timestamp inheritance.

2 of 2 metric rows shown. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 7 / 8

Chronological evaluation

No retained evidence applies to this topic in the declared scope.

0 of 0 metric rows shown. Presentation limits do not change analytical coverage.

Methodology and coverage

Evidence / source	Measurement or rule	Retained result
method	schema	1.4
method	methodology	temporal_reliability_v1
method	execution status	completed
method	aggregate scope	Every eligible declared record in required stages; no silent sampling.
method	search scope [0] / selected measurements	["MT_001","MT_002","MT_003","MT_004","MT_005","MT_006","MT_007","MT_008"]
method	search scope [0] / unsearched scope	["MT_009","MT_010","MT_011","MT_012","MT_013","MT_014","MT_015","MT_016","MT_017","MT_018","MT_019","MT_020","MT_021","MT_022","MT_023","MT_024","MT_025","MT_026","MT_027","MT_028","MT_029","MT_030","MT_031","MT_032","MT_033","MT_034","MT_035","MT_036","MT_037","MT_038","MT_039","MT_040","MT_041","MT_042","MT_043","MT_044","MT_045","MT_046","MT_… [retained JSON contains full detail]
method	search scope [0] / work observation visits	61151616
method	search scope [0] / comparisons	672
method	search scope [0] / selection	Requested measurements first, then declaration order. Detail retention can miss behavior-only exceptions.
method	search scope [0] / defaults validated universal defect criteria	No
method	internal detail omissions / electricity:entity histories	270
method	internal detail omissions / electricity:series effects	68
method	timezone / package	tzdata
method	timezone / version	2026.5
method	phase seconds / electricity:ingestion	53.78585
method	phase seconds / electricity:normalization	60.85567
method	phase seconds / required diagnostics	52.98135
method	phase seconds / availability history evaluation	0.0001065
method	phase seconds / behavior	15.73219
method	provisional limits / max cells	100000000

20 of 35 metric rows shown. 15 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Decoded-batch controls bound accepted batches and application retention. Native pages, dictionaries and conversion buffers can allocate before checks. Estimates and workload limits are provisional engineering controls. RSS monitoring can overshoot between samples.

Analytical scope left unsearched, retained-detail omissions, and presentation omissions are separate. No independent-row significance testing, model fitting, split generation, automatic repair or universal temporal score is performed.

Retained witnesses and selected detail

100 of 240 retained detail rows available here. Print shows up to 20 rows; JSON preserves every retained record. These presentation omissions do not change searched scope.

Evidence	Detail	Witness / selected population
t5	retained histories	{"entity":[],"measurement":"MT_001","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_002","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_003","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_004","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_005","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_006","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_007","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_008","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_009","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_010","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_011","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_012","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_013","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_014","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_015","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}

Evidence	Detail	Witness / selected population
t5	retained histories	{"entity":[],"measurement":"MT_016","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_017","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_018","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_019","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}
t5	retained histories	{"entity":[],"measurement":"MT_020","observations":140256,"cells":140256,"first_tick":"1293840900000000000","last_tick":"1420070400000000000","span_ticks":"126229500000000000"}